Sensor Placement Benchmark — California 2021, Uniform Coverage Kernel

Wildfire Drone Routing Project

2026-03-13

1. Introduction

This report presents sensor and drone placement results for the California 2021 wildfire dataset at three budget levels ($20M, $100M, $500M), using a **uniform coverage kernel** in the ILP formulation.

Dataset: 931 valid fires from the 2021 USFS California wildfire records, filtered for known date/time and valid WFPI data coverage.

Risk map: Pyrologix Ignition Probability (static, trained on 2006–2020 data, no data leakage for 2021).

Script: `run_benchmark_california2021_yearly.py`

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2. Coverage Kernel

2.1 Problem with the previous kernel

The original placement ILP used a kernel derived from a random walk diffusion process. This kernel had two issues:

1. **Normalization bug:** The kernel was divided by the origin value, inflating weights so they summed to roughly the number of reachable cells (~49) instead of the patrol budget (~7). This caused the ILP to assign unreasonably many drones per station (up to 58–76) to push corner cells to full coverage.
2. **Wrong model:** A random walk models independent, uncoordinated drones. In practice, the routing stage (TOP / MaxCoverage) coordinates drones to cover non-overlapping regions, eliminating the diminishing returns of independent walks.

2.2 Uniform kernel

The corrected kernel models coordinated drone patrol:

- **Reachable zone:** $B \times B$ cells, where $B = \text{max\_battery\_time} = 7$ (L_∞ distance $\leq \lfloor B/2 \rfloor = 3$ from the station).
- **Weight per cell:** $w = 1/B = 1/7 \approx 0.143$ for all 49 reachable cells.

- **Interpretation:** Each drone visits $B = 7$ distinct cells per charge cycle. With coordinated routing, B drones saturate the full zone ($B \times w \times B^2/B^2 = 1$).
- **ILP constraint:** $\theta_i \leq \sum_j w \cdot nc_j, \quad \theta_i \leq 1$.

Properties:

- Sum of kernel weights = $B^2/B = B = 7$ (total patrol budget of one drone).
- Drones needed to saturate one station's zone = $B = 7$.

2.3 Bug fixes applied

Two additional bugs were identified and fixed in the ILP formulation:

1. **Masked cells in objective (fixed):** The risk map contained non-zero values at masked (infeasible) cells — snow, urban, non-burnable areas. Since the objective summed $\text{risk}_i \cdot \theta_i$ over all $261 \times 161 = 42,021$ cells, 57.6% of the total risk came from infeasible cells. This distorted the objective and wasted budget on coverage of areas where no fires occur.

   Fix: `static_map .*= mask` — zero out risk for masked cells before building the ILP objective.

2. **Pre-filtering too aggressive (fixed):** A hardcoded top-20% candidate threshold restricted station candidates to ~2,238 cells, whose 7×7 zones could only reach 4,279 feasible cells (38% of 11,188). At $100M this forced 181 stations into an area needing only ~88, causing 49.1% coverage overlap waste and only 39.6% of feasible cells covered.

   Fix: `candidate_percentile = 0.00` — keep all feasible cells as candidates (solve time remains under 20 seconds).

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3. Simulation Parameters

Parameter	Value
Grid	California 2021 WFPI/Pyrologix, 1309×805 (~1 km/cell)
Operational grid	261×161 (5×5 coverage blocks)
Drone speed	600 m/min
Coverage radius	2900 m (~2.9 km)
Battery (max)	1 h = 7 operational substeps
Transmission range	50 km
Scenarios benchmarked	100 random fires (seed=42) from 931 valid
Placement strategy	`SensorPlacementMaxCoverageGaussianTimeMaskedBudget`
Solver	Gurobi (academic licence, 600 s time limit)
Candidate filtering	None (all 11,188 feasible cells)

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4. Routing Strategies

Two routing strategies are benchmarked on each placement.

Strategy 1 — `DroneRoutingTOPMasked` (TOP) PSO-based Team Orienteering Problem solver.

Parameter	Value
Reevaluation step	5 substeps
Optimization horizon	10 substeps
Time limit per call	60 s
Burn-map handling	static Pyrologix (tiled internally)
Visited-cell suppression	`reset_time = 2 × max_battery_time = 14` substeps

Strategy 2 — `DroneRoutingMaxCoverageGrowingMasked` (MaxCov) Greedy max-coverage routing with growing visited-cell suppression.

Parameter	Value
Reevaluation step	5 substeps
Optimization horizon	10 substeps
Time limit per call	60 s
Burn-map handling	static Pyrologix (tiled internally)
Visited-cell suppression	growing (cells visited during current horizon suppressed)

Both strategies use `burnmap_type="static"`: the single-frame Pyrologix map `(1, 261, 161)` is tiled internally to the full horizon length. Because the map never changes between scenarios, all clusters share a single routing log per strategy (`log_key="pyrologix"`).

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5. Budget Comparison

5.1 Allocation summary

	$20M	**$100M**	**$500M**
Ground sensors	0	0	2187
Charging stations	40	200	375
Drones	280	1400	2379
Budget used	$20.00M	$100.00M	$393.90M
MIP gap	0.0%	0.01%	0.0%
Solve time	2.4 s	19.6 s	2.6 s
Clusters	9	1	1

All three placements are **provably optimal** (MIP gap $\leq$ 0.01%).

5.2 Drone allocation

	\$20M	**\$100M**	**\$500M**
Stations with 7 drones	40 (all)	200 (all)	316
Stations with 0 drones	0	0	10
Stations with 1–6 drones	0	0	49

At \$20M and \$100M, the ILP assigns exactly 7 drones to every station (full zone saturation). At \$500M, the budget saturates — not all stations need 7 drones because the remaining cells are already covered by ground sensors.

5.3 Coverage efficiency (100M, before vs. after fix)

Metric	Before (top-20% filter)	After (no filter)
Stations	181	200
Feasible cells covered	4,436 (39.6%)	9,541 (85.3%)
Overlap waste	49.1%	0.4%
Objective achieved	48.1% of max	89.9% of max
Min station spacing (L_∞)	1	5

The fix eliminates nearly all overlap waste and more than doubles the number of feasible cells covered.

5.4 Key observations

\$20M: The optimizer allocates the full budget to stations (40) and drones (280), with zero ground sensors. At this budget level, drone coverage is more cost-effective than ground sensors. The 9 clusters target the highest-risk areas.

\$100M: Again, no ground sensors — all budget goes to 200 stations with 1,400 drones. The stations space themselves at $L_\infty \geq 5$ apart, covering 85.3% of all feasible cells with near-zero overlap. All stations merge into one connected cluster.

\$500M: The optimizer saturates at \$393.9M — it cannot usefully spend the remaining \$106.1M because all worthwhile placements are exhausted. 2,187 ground sensors blanket the interior of the covered zone, providing guaranteed single-cell detection. 375 stations with 2,379 drones cover virtually all burnable area.

6. Cluster Structure

6.1 Cluster counts

	\$20M	**\$100M**	**\$500M**
Singleton clusters	variable	0	0
Multi-station clusters	variable	1	1
Total	**9**	**1**	**1**

A fire scenario is **discoverable** if its ignition point (in operational space) falls within L_∞ distance ≤ 3 of at least one charging station (one-way reach on a single charge).

7. Placement Maps

7.1 $20M Budget

Data scale (1 km/cell)

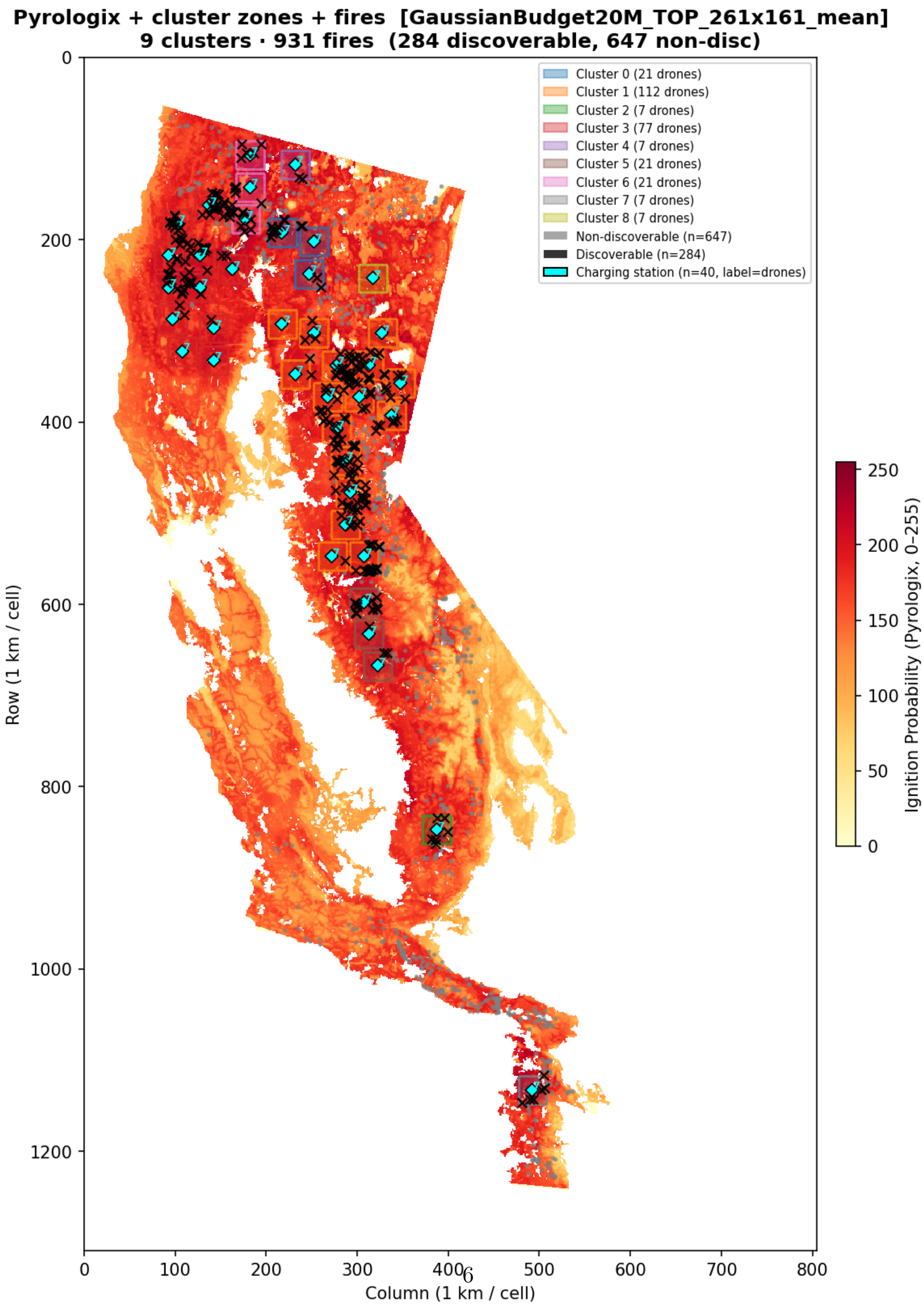

Figure 1: 20M placement — data scale. 9 clusters, 40 stations, 280 drones, 0 ground sensors. Cluster

Operational scale (5 km/cell)

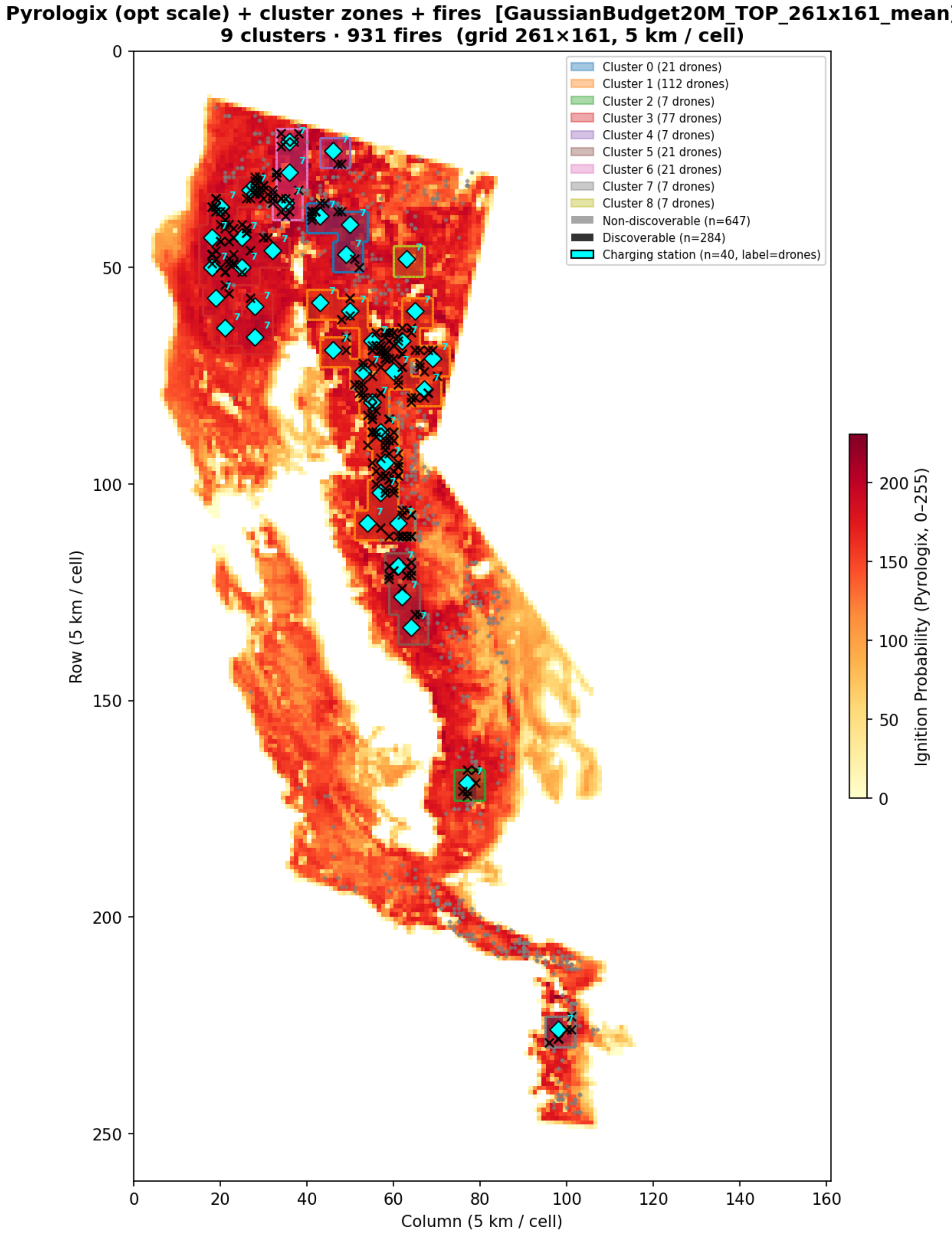

Figure 2: 20M placement — operational scale.

7.2 $100M Budget

Data scale (1 km/cell)

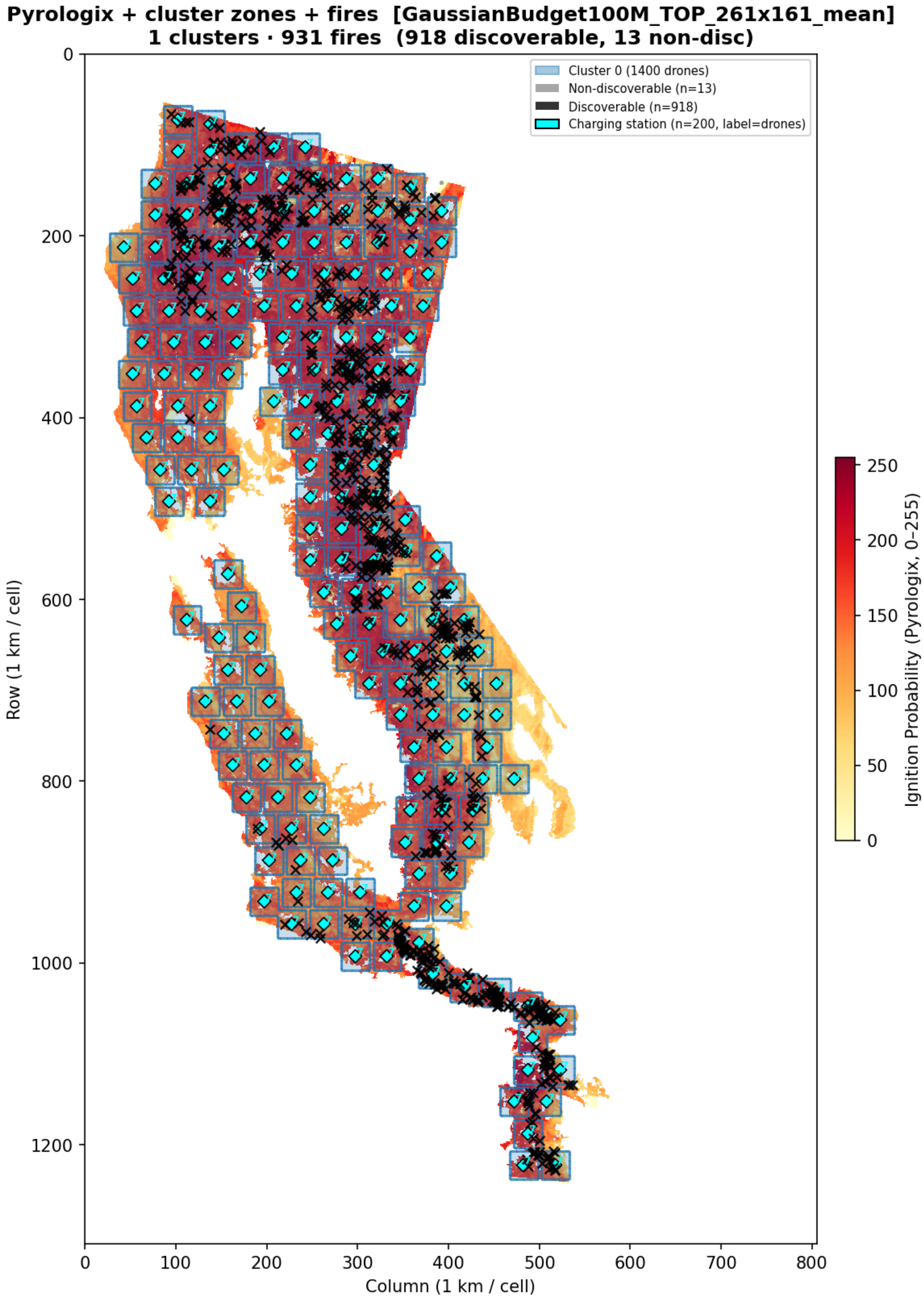

Figure 3: 100M placement — data scale. 1 cluster, 200 stations, 1400 drones, 0 ground sensors. Stations spaced at $L_\infty \geq 5$, covering 85% of feasible cells with near-zero overlap.

Operational scale (5 km/cell)

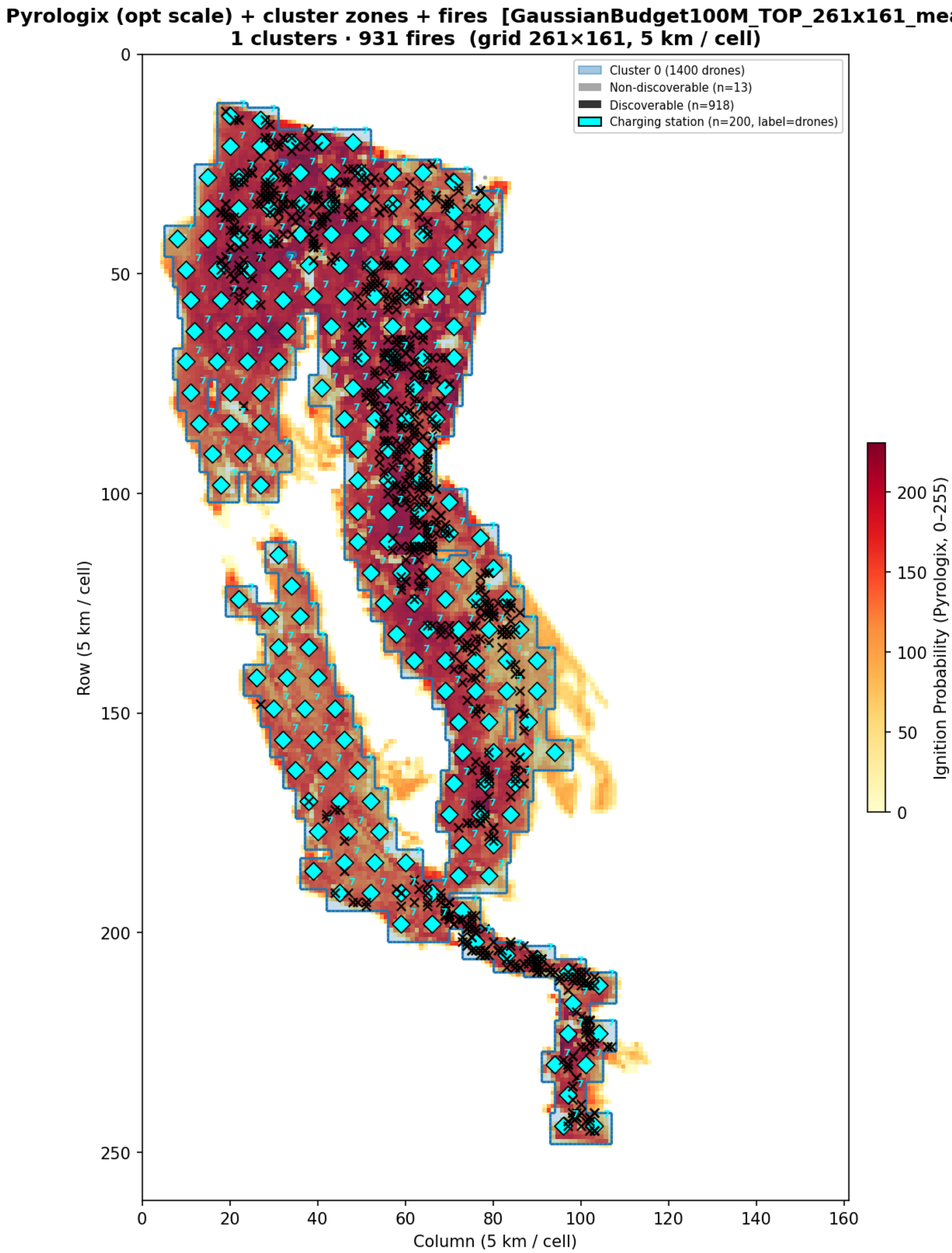

Figure 4: 100M placement — operational scale.

7.3 $500M Budget

Data scale (1 km/cell)

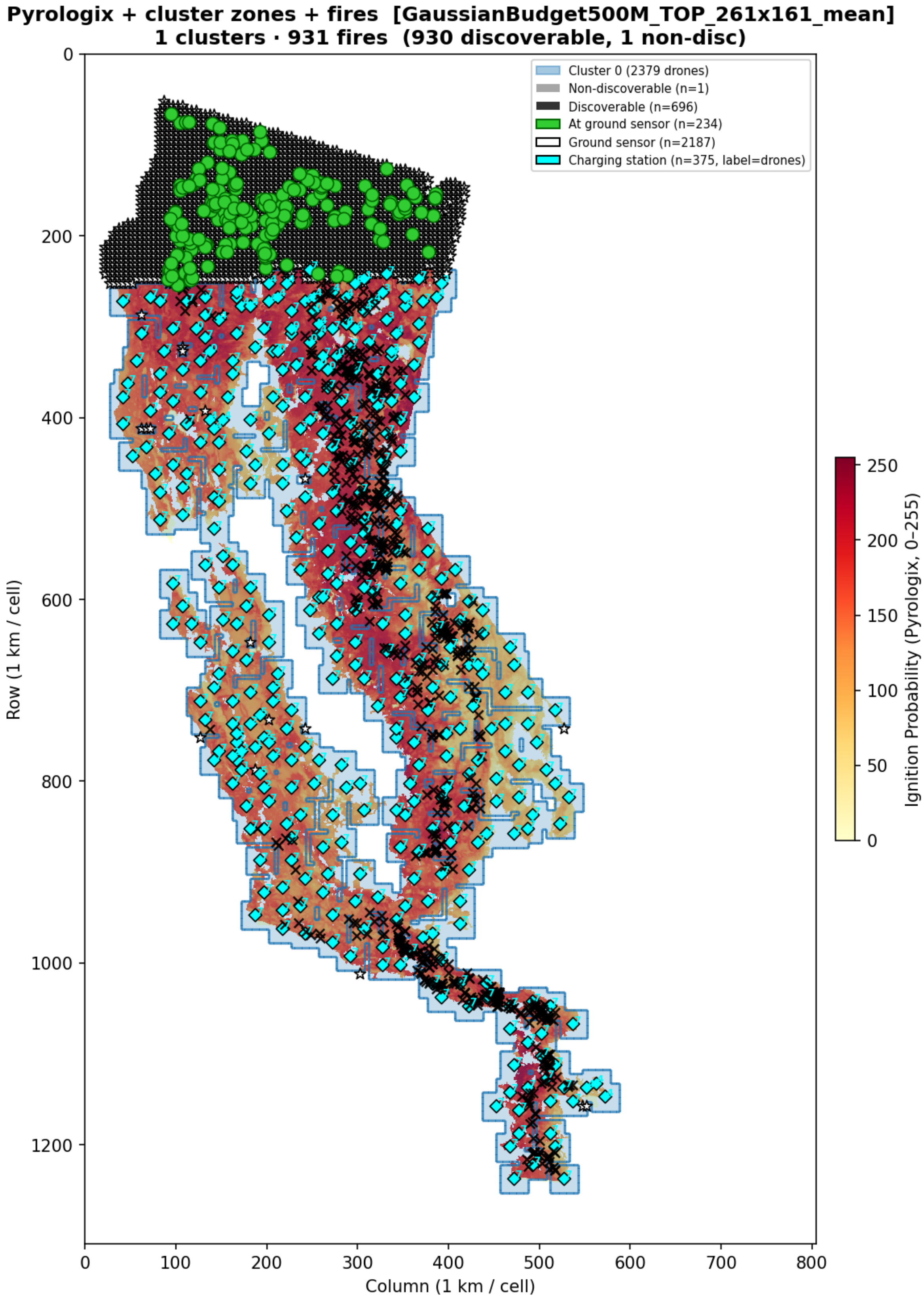

Figure 5: 500M placement — data scale. 1 cluster, 375 stations, 2379 drones, 2187 ground sensors. Budget saturates at $393.9M.

Operational scale (5 km/cell)

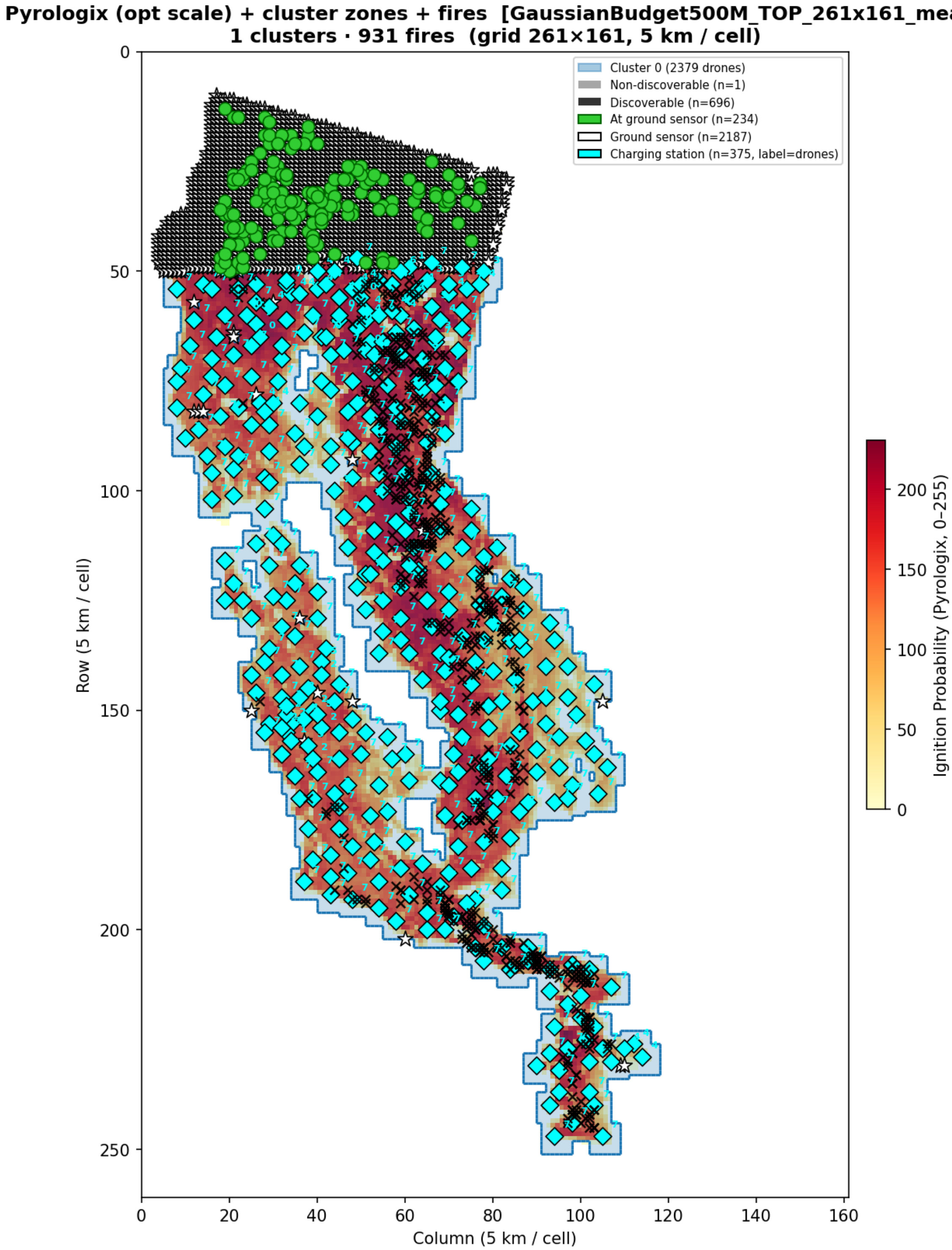

Figure 6: 500M placement — operational scale.

8. Detection Results

Routing benchmarks (TOP and MaxCoverage strategies) are pending for all three budget levels. With the corrected ILP, station spacing is now ≥ 5 cells (at \$100M), producing well-separated clusters with exactly 7 drones per station. This should yield efficient PSO routing with no bottleneck from oversized clusters.

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9. How to Reproduce

Sensor placement only:

```
python-jl run_benchmark_california2021_yearly.py --sensor-only --budget 20
python-jl run_benchmark_california2021_yearly.py --sensor-only --budget 100
python-jl run_benchmark_california2021_yearly.py --sensor-only --budget 500
```

Generate placement plots:

```
python visualize_sensor_placement_2021.py \
  California2021Dataset/logs/sensor_alloc_GaussianBudget20M_TOP_261x161_mean.json \
  --scale both
python visualize_sensor_placement_2021.py \
  California2021Dataset/logs/sensor_alloc_GaussianBudget100M_TOP_261x161_mean.json \
  --scale both --tag _100M
python visualize_sensor_placement_2021.py \
  California2021Dataset/logs/sensor_alloc_GaussianBudget500M_TOP_261x161_mean.json \
  --scale both --tag _500M
```

Full benchmark (placement + routing):

```
python-jl run_benchmark_california2021_yearly.py --budget 20
python-jl run_benchmark_california2021_yearly.py --budget 100
python-jl run_benchmark_california2021_yearly.py --budget 500
```